Assignment 1: AI System Specification

PEAS Framework & Task Environment Analysis

Course: Introduction to Artificial Intelligence | Option C: Agricultural Monitoring Drone

Task 1: PEAS Specification

System Overview

The selected agent is an **Autonomous Agricultural Monitoring Drone** designed for early-stage crop disease detection across large farming fields. The drone operates by autonomously navigating pre-planned flight paths, capturing multi-spectral imagery, and applying on-board AI inference to identify diseased zones — alerting farm managers in real time.

PEAS Breakdown

Performance Measure	1. **Disease Detection Accuracy:** Percentage of correctly identified diseased crop zones (target >95% precision, >90% recall). 2. **Field Coverage Rate:** Percentage of total farmland surveyed per mission without gaps. 3. **False Positive Rate:** Ratio of healthy zones incorrectly flagged as diseased (target <3%). 4. **Mission Completion Time:** Total time to survey a standard 10-hectare field per battery cycle. 5. **Alert Latency:** Time from detection of disease to delivery of geo-tagged report to farm manager.
Environment	**Physical space:** Open agricultural fields, orchards, and greenhouses with varying crop row layouts. **Weather conditions:** Variable wind speed (<15 m/s operational limit), humidity, and ambient light. **Terrain:** Flat to mildly undulating land; potential for GPS signal interference near tall structures. **Biological context:** Crops at various growth stages, pest presence, and soil moisture variation.
Actuators	**Quadrotor propulsion system** — 4 independently controlled brushless DC motors for navigation, altitude, and hover. **Gimbal-stabilized camera mount** — 3-axis gimbal for steady nadir and oblique imaging. **Wireless telemetry transmitter** — 4G/LTE and Wi-Fi module for real-time data upload to cloud dashboard. **Audio/visual alert beacon** — on-board LED indicator and buzzer for low-battery and fault signaling.

Sensors	**High-resolution RGB camera (20 MP)** — captures true-color imagery for visual disease symptom identification. **Multispectral sensor (5-band: Red, Green, Blue, Red-Edge, Near-Infrared)** — computes NDVI and chlorophyll indices to detect sub-visual stress. **Thermal infrared sensor (FLIR Lepton, 160×120)** — identifies temperature anomalies indicative of fungal or bacterial infections. **GPS/GNSS module (RTK-enabled, ±2 cm accuracy)** — provides precise geo-tagging of detected disease locations. **IMU (6-DOF accelerometer + gyroscope)** — stabilizes flight and compensates for wind disturbances. **LiDAR altimeter** — maintains consistent flight altitude above crop canopy for standardized image capture.

Task 2: Environment Classification

Observable	**Partially Observable**	Sensors capture only a narrow field-of-view per instant; sub-canopy regions, root zones, and areas obscured by cloud shadow remain unobservable during flight.
Agent Type	**Multi-agent**	In commercial deployments, a swarm of drones divides the field for parallel coverage, requiring coordinated path planning to avoid collisions and redundant scanning.
Determinism	**Stochastic**	Wind gusts, variable lighting conditions, and sensor noise introduce unpredictability, meaning the same actuator command does not always produce an identical flight trajectory or image quality.
Episode Type	**Sequential**	The drone's current scanning position and image history directly influence which zones to prioritize next; prior detections inform the adaptive re-routing of the flight path.
Dynamism	**Dynamic**	The environment changes continuously during operation — wind speed shifts, lighting transitions from direct sun to cloud cover, and crop canopy movement alter sensor readings in real time.
State Space	**Continuous**	The drone operates in a continuous 3D spatial domain; sensor readings (NDVI values, thermal gradients, GPS coordinates) are real-valued and not discretely enumerable.

Task 3: Critical Analysis

Q1. Greatest Technical Challenge

Among the six dimensions, the **Stochastic** nature of the environment poses the greatest technical challenge. Unlike deterministic systems where outcomes are predictable, the agricultural drone must contend with continuously changing wind patterns, shifting illumination, and sensor noise — all of which can degrade model performance unpredictably. A disease detection model trained under clear-sky, calm-wind conditions may produce unreliable NDVI readings under overcast or turbulent conditions, leading to missed detections or false alarms. Engineers must build robust probabilistic models, apply sensor fusion across multiple modalities (RGB + multispectral + thermal), and implement adaptive exposure control — significantly increasing system complexity compared to a deterministic counterpart.

Q2. Proposed Utility Function

A key trade-off in this system is between **Detection Thoroughness** (high recall, scanning slowly and repeatedly) and **Mission Efficiency** (covering maximum area per battery charge). The following utility function captures this balance:

$$U = \alpha \times \text{RecallScore} - \beta \times \text{MissionTime} - \gamma \times \text{FalsePositiveRate}$$

Where α weights detection recall, β penalizes excessive time usage, and γ penalizes false alarms. With default weights (α=*0.5,* β=*0.3,* γ=*0.2*), the drone balances speed and accuracy.

If α **is doubled** (e.g., from 0.5 to 1.0), the agent would strongly prioritize maximizing recall — it would fly lower, reduce speed for higher image resolution, re-scan suspicious zones multiple times, and accept longer mission durations to ensure no diseased area is missed. This behavioral shift is desirable during critical growth stages (e.g., pre-harvest) where undetected disease could cause catastrophic crop loss, but it would drain battery faster and reduce total field coverage per flight — a trade-off the farm manager must consciously accept.